TAMPINES JUNIOR COLLEGE

Preliminary Examinations 2008

CHEMISTRY (Higher 2) 9746/3

PAPER 3 Free Response

Tuesday **19 August 2008** 2 hours

Additional materials: Writing Paper
Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your name and Civics Group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, detach this cover sheet and fasten it together with your work.

Name : ______________________

CG : ________

Question	Marks
1	
2	
3	
4	
5	
TOTAL	/ 80

This document consists of **7** printed pages.

[Turn over

*Answer any **four** questions.*

1 (a) (i) Chlorine is an ***oxidising agent.*** Explain what is meant by the term in ***italics***.
Write a balanced equation for the reaction of chlorine with $SO_3^{2-}(aq)$ and use it to illustrate your answer.

(ii) Describe one simple chemical test to show that the expected oxidation had taken place.

[4]

(b) The standard reduction potential for hydrazine, N_2H_4 , in acidic solution is as shown:

$$N_2(g) + 5H^+(aq) + 4e^- \rightleftharpoons N_2H_5^+(aq) \qquad E^\ominus = -0.17 \text{ V}$$

By using this and any other necessary information from the *Data Booklet*, predict the outcome of the reaction of chlorine with hydrazine in acidic solution and write a balanced equation for the overall reaction.

[2]

(c) (i) Write down the form in which bromine (other than molecular Br_2) could acts as

- an electrophile, and
- a free radical.

(ii) Bromoethanoic acid, $BrCH_2COOH$, can be obtained by reaction of bromine gas with ethanoic acid. Name and outline the steps involved in the mechanism of this reaction.

[5]

(d) (i) Hydrogen fluoride, HF, in aqueous solution, is a much weaker acid than hydrogen chloride, HC*l*. Apart from the fact that H–F has a stronger bond energy than that of H–C*l*, suggest another reason why HF has a much lower K_a value than expected.

(ii) Calculate the pH of a 0.200 mol dm^{-3} aqueous solution of HF at 298K.
[K_a of HF = 5.62 x 10^{-4} mol dm^{-3} at 298K]

(iii) Calculate the pH of 100 cm^3 of a 0.200 mol dm^{-3} aqueous solution of HF after the addition of 50.0 cm^3 of a 0.100 mol dm^{-3} aqueous solution of sodium hydroxide.

(iv) F^- is the conjugate base of hydrofluoric acid, HF. When calcium fluoride, CaF_2, dissolves in water, it undergoes a hydrolysis reaction to form a weakly alkaline solution.
Write an ionic equation, with state symbols, for the hydrolysis reaction of fluoride ion in water.
Calculate the pH of a 0.0011 mol dm^{-3} solution of calcium fluoride.
[K_w = 1.00 x 10^{-14} mol^2 dm^{-6} at 298K]

[9]

[Total: 20]

2 **(a)** What do you understand by the terms ***rate constant*** and ***half-life*** ? [2]

(b) The hydrolysis of a bromoalkane, RBr, by aqueous NaOH proceeds as follows:

$$RBr + NaOH \longrightarrow ROH + NaBr$$

Experimental results show that the reaction is first order with respect to RBr and first order with respect to NaOH.

(i) Write the rate equation for the reaction and give the units of the rate constant, *k*.

(ii) A series of experiments was conducted and the following results were obtained:

Experiment	[RBr] / mol dm^{-3}	[NaOH] / mol dm^{-3}	half-life of RBr / min
1	x	x	40
2	$2x$	x	?
3	x	$4x$	?

State, with reasoning, what you would expect the half-life of the bromoalkane to be in Experiments 2 & 3.

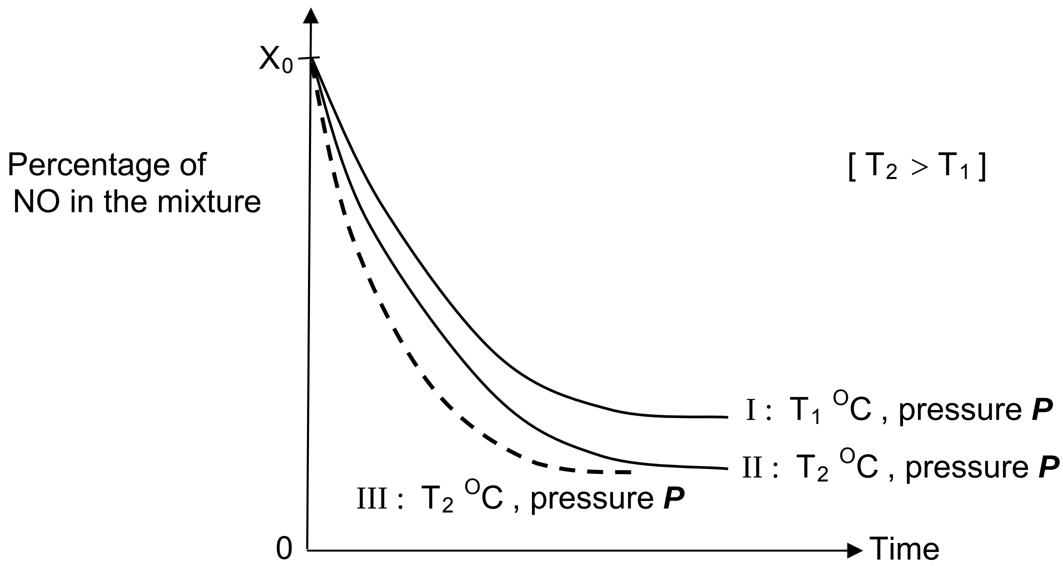
(iii) In view of the experimental results, suggest a mechanism for the hydrolysis of the bromoalkane by sodium hydroxide.

[7]

(c) Nitrosyl bromide, NOBr, can be obtained from nitric oxide (NO) and Br_2 according to the following equation.

$$2NO\,(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$$

Equimolar amounts, X_0 , of NO and Br_2 were mixed at a constant pressure ***P*** but at two different temperatures, T_1 ^{O}C and T_2 ^{O}C (where $T_2 > T_1$). The figure below shows the variation of the amount of NO remaining in the mixture with time for the two experiments.



(i) Why is the initial slope of curve II greater than that of curve I?

(ii) The percentage of NO in the mixture approaches a constant value. What conclusion about the reaction can be drawn from this?

(iii) What information may be obtained from the curves about the sign of ΔH (enthalpy change) of the forward reaction $2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$?

(iv) Suggest what single change was made to the experimental conditions that could have resulted in curve **III** being obtained. Explain your reasoning.

[6]

(d) Nitric oxide complexes with transition metals to form metal nitrosyls which have important biological functions.

(i) Draw a dot-and-cross diagram of nitric oxide.

(ii) Suggest the type of bond nitrogen oxide forms with the transition metal and state the feature it possesses that enables it to do so.

(iii) A medicinally important nitrosyl complex is the nitroprusside anion, $[Fe(CN)_5NO]^{2-}$. For this complex, determine the oxidation state of iron and give its electronic configuration.

[5]

[Total: 20]

3 **(a)** Methane is burnt in some electricity generating power stations. Commercial methane often contains hydrogen sulphide, H_2S, as an impurity. During the combustion of commercial methane, hydrogen sulphide reacts with oxygen to form sulphur dioxide.

(i) Write the equation for the reaction between hydrogen sulphide and oxygen.

(ii) A sample of commercial methane contains 0.005% by volume of hydrogen sulphide. Calculate the volume of sulphur dioxide produced when 1000 dm^3 commercial methane is burnt. All volumes are measured at room temperature and pressure.

(iii) One method of reducing the emission of sulphur dioxide from the waste gases is to treat the waste gases with powdered limestone, $CaCO_3$, producing calcium sulphite, $CaSO_3$. This is oxidised by air to form solid calcium sulphate, $CaSO_4$.
Write a balanced equation, with state symbols in each case, to show how the

I limestone reacts with sulphur dioxide,

II air oxidises calcium sulphite.

[4]

(b) Describe the shapes of, and state the bond angles in, the molecules of hydrogen sulphide, H_2S, and sulphur dioxide, SO_2.
Briefly explain your answers in terms of the number and types of electron pairs around the central atom. [4]

(c) The sulphur atom in SO_2 is joined to the oxygen atom by a ***σ-bond*** and a ***π-bond***.
Explain, with the aid of diagrams, what is meant by the terms in *italics*. [3]

(d) Carbon dioxide sublimes at –78 °C whereas the boiling point of sulphur dioxide is –10°C. Explain this observation in terms of bonding and structure. [3]

(e) The reaction between peroxodisulphate (VII) , $S_2O_8^{2-}$, and iodide ions is represented by the following equation:

$$2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

The reaction is catalysed by iron(III) ions.

(i) Explain what is meant by the activation energy of a chemical reaction and use the concept to explain, in general terms, the effect of a catalyst on the rate of a chemical reaction.

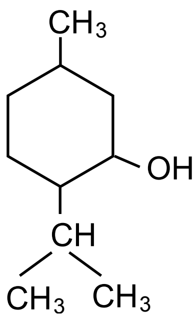
(ii) Explain, with the aid of equations, how the iron(III) ion carries out its role as a catalyst in the reaction between peroxodisulphate and iodide ions.

[6]

[Total: 20]

4 **(a)** Compound **Y** is generally known as Menthol. It can be found naturally in peppermint oil. It has many commercial uses, for example in cough drops and nasal inhalers. Menthol, $C_{10}H_{20}O$, has the structural formula given below. It is a substituted cyclohexanol whose systematic name is 5-methyl-2-(1-methylethyl)cyclohexanol.

Y



(i) Copy down the structural formula of **Y** and, by means of an asterisk (*), identify the chiral centre(s) in the formula you have drawn.
How many stereoisomers would you expect menthol to have? Explain your answer.

(ii) The molecule of menthol is ***optically active***. Briefly explain what you understand by the term ***optically active***.

[3]

(b) The following reactions involve transforming menthol, **Y**, into **S**, **T**, **U** and **V**.

$$\begin{tikzcd}
\underset{\textbf{S}}{C_{10}H_{19}Cl} & \underset{\textbf{Y}}{C_{10}H_{20}O} \arrow[l, "HCl(g)"'] \arrow[r, "\text{excess } H_2SO_4(l)", "180\ ^{o}C"'] \arrow[d, "\text{hot } K_2Cr_2O_7(aq)/H^+"] & \underset{\textbf{T}}{C_{10}H_{18}} \arrow[d, "\text{hot } KMnO_4(aq)/H^+"] \\
 & \underset{\textbf{U}}{C_{10}H_{18}O} & \underset{\textbf{V}}{C_{10}H_{18}O_3}
\end{tikzcd}$$

(i) Identify **S, T, U** and **V** by writing their structural formulae.

(ii) State the type of reaction that occurs when **Y** is transformed into **T.**

(iii) **T** can also be prepared from **S**. Suggest reagents and conditions for this conversion.

(iv) Suggest a simple chemical test to identify the functional group present in **U,** giving reagents, conditions and the observation you would expect from the reaction**.**

[8]

(c) Menthol is found naturally in peppermint oil. A sample of peppermint oil contains 40 mg of extracted menthol in a volume of 320 cm^3.
Calculate the concentration of menthol in g dm^{-3}. [1]

(d) Explain each of the following observations as fully as you can.

(i) The boiling points of menthol (Mr =156) and n-Undecane (Mr =156), $CH_3(CH_2)_9CH_3$, are 212 ^{o}C and 196 ^{o}C respectively.

(ii) Menthol is less soluble in water than ethanol.

[5]

(e) The enthalpy change of combustion of menthol is –6310 kJ mol^{-1}. What mass of menthol needs to be combusted in order to produce enough energy to boil 500 g of water initially at 25^{o}C? (Assume the process to be 75% efficient.) [3]

[Total: 20]

[continued on next page

5 (a) When methylbenzene is nitrated at slightly elevated temperatures by a mixture of concentrated nitric acid and concentrated sulphuric acid, the product consists largely of two isomers, **P** and **Q**, of formula $C_7H_7NO_2$.

In contrast, phenol is nitrated by dilute aqueous nitric acid, even at room temperature, and gives rise to two isomers, **R** and **S**, of formula $C_6H_5NO_3$.

(i) Draw the structural formulae of the isomer-pair of **P** and **Q**, and the isomer-pair of **R** and **S.**

(ii) State and briefly outline with the aid of equations, the mechanism of the nitration of phenol by dilute aqueous nitric acid.

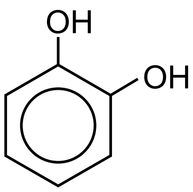
(iii) What can you deduce from the above two nitration reactions in terms of the reactivity of the aromatic rings present in both methylbenzene and phenol? Explain your answer.

(iv) Suggest which isomer, **R** or **S**, is the major product in the nitration of phenol. Explain your answer.
[Note that the ratio of the major to the minor product is 15:1.]

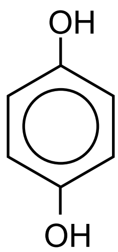
(v) Would you expect the nitrated phenol to be a stronger or weaker acid than phenol? Explain your answer.

[11]

(b)



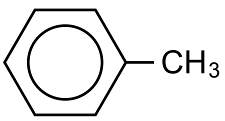
catechol

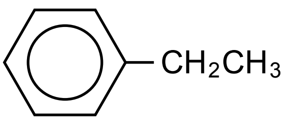


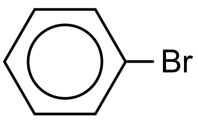
hydroquinone

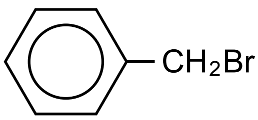
Catechol has a lower melting point (104 °C) than hydroquinone (169 °C). Explain this observation. [3]

(c) For **each** of the following pairs of compounds, describe **one** simple chemical test which would enable you to distinguish between them.
State clearly how each compound behaves in the test.
Your descriptions should include the reagents and conditions used, observations and the displayed formula of the organic products formed.

(i) 

and 

(ii) 

and 

[6]

[Total: 20]

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